



## **PART [B]: GENERAL PROGRAMS**

### **(6) Electrical Power Engineering Program**

برنامج هندسة القوى الكهربائية



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### (6) Electrical Power Engineering Program (EPE)

برنامج هندسة القوى الكهربائية

#### VISION رؤية البرنامج

Pioneering education, community service and qualification for scientific research in the field of electrical power engineering locally, regionally and globally.

ريادة التعليم وخدمة المجتمع والتأهيل للبحث العلمي في مجال هندسة القوى الكهربائية محليا وإقليميا وعالميا.

#### MISSION رسالة البرنامج

The program seeks to reach a high competitive level by preparing a distinguished and innovative engineer in the field of electrical power engineering and its applications, who is competent in applying advanced scientific knowledge, communication skills and tools and adhering to professional ethics by keeping pace with market needs, achieving sustainable economic development and community service, and armed with the skills to conduct scientific research, and urge students towards the foundations of entrepreneurship.

يسعى البرنامج للوصول الى مستوى تنافسي عال من خلال إعداد مهندس متميز ومبتكر في مجال هندسة القوى الكهربائية وتطبيقاتها، يكون قادرا على استخدام المعرفة العلمية المتقدمة ومهارات الاتصال وأدواته و متمسكا بأخلاقيات المهنة وذلك بمواكبة احتياجات السوق وتحقيق التنمية الاقتصادية المستدامة وخدمة المجتمع، ومسلحا بمهارات القيام بالبحث العلمي. كما يحث البرنامج الطلبة على الاهتمام بأسس ريادة الأعمال.



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### GRADUATE ATTRIBUTES مواصفات الخريج

The graduate should be able to:

1. Design, operate and maintain systems used to generate, transmit, distribute and control electrical energy.
2. Design and develop devices used in the electrical industrial fields.
3. Acquire considerable knowledge of project management, logistics, procurement and supply chain management.
4. Be innovative and entrepreneurial.
5. Aware of the impact of various electrical power engineering activities on society and the environment.
6. Gain leadership competencies as well as social and emotional intelligence.

1. القدرة على تصميم وتشغيل وصيانة الأنظمة المستخدمة لتوليد الطاقة الكهربائية ونقلها وتوزيعها والتحكم فيها.
2. القدرة على تصميم وتطوير الأجهزة المستخدمة في المجالات الصناعية الكهربائية.
3. المعرفة المعتمدة بإدارة المشاريع واللوجستيات والمشتريات وإدارة سلسلة التوريد.
4. أن يكون مبتكرا ورائدا للأعمال.
5. الدراية بتأثير أنشطة هندسة الطاقة الكهربائية المختلفة على المجتمع والبيئة.
6. اكتساب الكفاءات القيادية وكذلك الذكاء الاجتماعي والعاطفي.

### PROGRAM BENCHMARK مرجعية البرنامج

يستند تصميم البرنامج للمعايير الأكاديمية القومية للعام 2018 بالإضافة إلى المعايير المرجعية **ABET** والمعايير المرجعية الكندية **CEAB**.

| NARS 2018 | LEVEL A                   | LEVEL B         | LEVEL C            |
|-----------|---------------------------|-----------------|--------------------|
|           | Totally Adopted<br>P. A11 | Totally Adopted | Adopted- See below |



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### **In addition to the Competencies for All Engineering Programs the BASIC ELECTRICAL Engineering (LEVEL B) graduate must be able to:**

1. Select, model and analyze electrical power systems applicable to the specific discipline by applying the concepts of: generation, transmission and distribution of electrical power systems.
2. Design, model and analyze an electrical/electronic/digital system or component for a specific application; and identify the tools required to optimize this design.
3. Design and implement: elements, modules, sub-systems or systems in electrical/electronic/digital engineering using technological and professional tools.
4. Estimate and measure the performance of an electrical/electronic/digital system and circuit under specific input excitation, and evaluate its suitability for a specific application.
5. Adopt suitable national and international standards and codes to: design, build, operate, inspect and maintain electrical/electronic/digital equipment, systems and services.

### **LEVEL C:**

### **In addition to the Competencies for BASIC ELECTRICAL Engineering graduate, EPE graduate must be able to:**

1. Design, commission, operate and maintain systems used for the generation, transmission, distribution and control of electrical energy.
2. Design and develop the apparatus employed by the electrical power industry.
3. Acquire good knowledge of project management, logistics, procurement, and supply chain management.
4. Be innovative and motivated by entrepreneurship.
5. Be aware of the impact of different electrical power engineering activities on the society and environment.
6. Acquire leadership competences as well as social and emotional intelligence.



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### PROGRAM REQUIREMENTS متطلبات البرنامج

\*\*\* المقرر التخصصي للطلاب المتفوق: GENG121 Selected Topics 3:

| Category                         |                  | No. of courses | Course Credit Hour | Total Credit Hours |
|----------------------------------|------------------|----------------|--------------------|--------------------|
| Discipline Requirements (DR)     | core/ compulsory | 1              | 4                  | 4                  |
|                                  |                  | 14             | 3                  | 42                 |
|                                  |                  | 6              | 2                  | 12                 |
|                                  |                  | 1              | 1                  | 1                  |
|                                  | Elective         | 0              | 0                  | 0                  |
|                                  |                  | 0              | 0                  | 0                  |
| Total DR courses                 |                  | 22             |                    | 59                 |
| Program Requirement (PR)         | core/ compulsory | 8              | 3                  | 24                 |
|                                  |                  | 1              | 2                  | 2                  |
|                                  | Elective         | 6              | 3                  | 18                 |
|                                  |                  | 0              | 2                  | 0                  |
| Total PR courses                 |                  | 15             |                    | 44                 |
| Total Elective courses (DR & PR) |                  | 6              | 3                  | 18                 |



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### ▪ Discipline Requirements (DR) core/compulsory courses list

| Code     | Name                                              | Credit Hours | Pre-requisite          |
|----------|---------------------------------------------------|--------------|------------------------|
| EMCG101  | Dynamics of Rigid bodies                          | 3            | EMCG001                |
| CVEG118  | Civil Engineering                                 | 2            | EMCG001                |
| CMPEG218 | Data Structures                                   | 2            | AMTG005                |
| EECG118  | Electronics                                       | 3            | PHYG001                |
| EECG318  | Electrical Communications Systems                 | 2            | MTHG003 + EPEG204      |
| EPEG101  | Electrical Circuits (1)                           | 3            | PHYG002+MTHG003        |
| EPEG102  | Basics of Energy Conversion                       | 2            | PHYG002+MTHG003        |
| EPEG103  | Computer Applications in Electrical Power Systems | 3            | AMTG005                |
| EPEG104  | Electromagnetic Fields                            | 4            | PHYG002+MTHG104        |
| EPEG105  | Electrical Circuits (2)                           | 3            | MTHG104+EP EG101       |
| EPEG106  | Logic Circuits and Microprocessors                | 3            | EPEG103+EE CG118       |
| EPEG201  | Power Systems (1)                                 | 3            | EPEG105+EP EG202       |
| EPEG202  | Electrical Machines (1)                           | 3            | EPEG105+EP EG104       |
| EPEG203  | Electric and Electronic Measurements              | 3            | EPEG101+EE CG118       |
| EPEG204  | Signals and Systems                               | 3            | MTHG104+ EPEG101       |
| EPEG207  | Automatic Control Systems                         | 3            | EPEG204                |
| EPEG280  | Summer Internship                                 | 1            | Freshman Level Courses |
| EPEG306  | Power Electronics (2)                             | 3            | EPEG302                |
| EPEG307  | Digital Control Systems                           | 3            | EPEG207                |



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| Code         | Name                          | Credit Hours | Pre-requisite                                  |
|--------------|-------------------------------|--------------|------------------------------------------------|
| EPEG480      | Graduation Project 1          | 1            | 110 Credit hours + All Sophomore Level courses |
| EPEG481      | Graduation Project 2          | 3            | EPEG480+ AA Approval                           |
| INTG118      | Electric Materials            | 2            | PHYG002                                        |
| MDPG102      | Elements of Mechanical Design | 2            | EMCG101                                        |
| MEPG173      | Engineering Thermodynamics    | 3            | PHYG001                                        |
| <b>Total</b> |                               | <b>59</b>    |                                                |

### ▪ Program Requirements (PR) core/compulsory courses list

| Code         | Name                                         | Credit Hours | Pre-requisite    |
|--------------|----------------------------------------------|--------------|------------------|
| EPEG205      | Electrical Machines (2)                      | 3            | EPEG202          |
| EPEG206      | Electrical Power Generation                  | 3            | EPEG101+ME PG173 |
| EPEG302      | Power Electronics (1)                        | 3            | EECG118+EP E105  |
| EPEG303      | Electrical Machines (3)                      | 3            | EPEG205          |
| EPEG304      | Switchgear and Protection Systems            | 3            | EPEG201+EP EG203 |
| EPEG305      | High Voltage Engineering                     | 3            | EPEG203          |
| EPEG301      | Power Systems (2)                            | 3            | EPEG201+MT HG110 |
| EPEG401      | Electrical Energy Utilization and Management | 3            | EPEG205+EP EG306 |
| EPEG402      | Electrical Installations                     | 2            | EPEG304+EP EG205 |
| <b>Total</b> |                                              | <b>26</b>    |                  |



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### ▪ Program Requirements (PR) elective courses list

| Code                      | Name                                                      | Credit Hours | Pre-requisite         |
|---------------------------|-----------------------------------------------------------|--------------|-----------------------|
| <b>ELECTIVE (EPEG4xx)</b> |                                                           |              |                       |
| EPEG411                   | Electrical Power System Planning                          | 3            | EPEG301 + AA Approval |
| EPEG412                   | Advanced Studies in Electrical Power Systems              | 3            | EPEG301 + AA Approval |
| EPEG413                   | Effect of Power Quality on Electrical Utilization Systems | 3            | EPEG301 + AA Approval |
| EPEG414                   | Electrical Substations                                    | 3            | EPEG301 + AA Approval |
| EPEG415                   | Renewable Energy Systems                                  | 3            | EPEG301 + AA Approval |
| EPEG416                   | Renewable Energy Sources Grid Connected Systems           | 3            | EPEG301 + AA Approval |
| EPEG417                   | Demand Side Management                                    | 3            | EPEG301 + AA Approval |
| EPEG418                   | Cybersecurity Applied to the National Grid                | 3            | EPEG301 + AA Approval |
| EPEG419                   | Data Communication for Instrumentation and Control        | 3            | EPEG307 + AA Approval |
| EPEG420                   | Special Topics in Automatic Control                       | 3            | EPEG307 + AA Approval |
| EPEG421                   | Intelligent Control                                       | 3            | EPEG307 + AA Approval |
| EPEG422                   | Industrial Control System Components                      | 3            | EPEG307 + AA Approval |
| EPEG423                   | Computer Controlled Systems                               | 3            | EPEG307 + AA Approval |
| EPEG424                   | Selected Topics in Electrical Machines and Transformers   | 3            | EPEG303 + AA Approval |
| EPEG425                   | Applications of Computers in Electrical Machine Design    | 3            | EPEG303 + AA Approval |
| EPEG426                   | Electric Drive Systems                                    | 3            | EPEG303 + AA Approval |





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| Code    | Name                                                       | Credit Hours | Pre-requisite         |
|---------|------------------------------------------------------------|--------------|-----------------------|
| EPEG427 | Advanced Drive Systems                                     | 3            | EPEG426 + AA Approval |
| EPEG428 | Electric Traction and Mobility Systems                     | 3            | EPEG303 + AA Approval |
| EPEG429 | Electrical Generators for Renewable Energy Applications    | 3            | EPEG303 + AA Approval |
| EPEG430 | Selected topics in high voltage engineering                | 3            | EPEG305 + AA Approval |
| EPEG431 | High voltage engineering applications                      | 3            | EPEG305 + AA Approval |
| EPEG432 | High Voltage Direct Current Transmission                   | 3            | EPEG305 + AA Approval |
| EPEG433 | Advanced Topics in protection Systems                      | 3            | EPEG304 + AA Approval |
| EPEG434 | DC Circuit Breakers                                        | 3            | EPEG304 + AA Approval |
| EPEG435 | Power Electronics Applications in Renewable Energy Systems | 3            | EPEG306 + AA Approval |
| EPEG436 | Power Electronics Applications in Power Systems            | 3            | EPEG306 + AA Approval |
| EPEG437 | Embedded Systems Applications in Power Electronics         | 3            | EPEG306 + AA Approval |
| EPEG438 | Power Electronics for Electric Vehicles                    | 3            | EPEG306 + AA Approval |
| EPEG439 | Operations Research                                        | 3            | AA Approval           |
| EPEG440 | Electrical Project Management                              | 3            | AA Approval           |



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### Program Plan - Courses After Freshman – According to Level Sequence

#### Level 100 Courses

| Code    | Name                                              | Credit Hours | Contact Hours |         |          |     |      |          |          |       |
|---------|---------------------------------------------------|--------------|---------------|---------|----------|-----|------|----------|----------|-------|
|         |                                                   |              | Lec           | Tut (2) | App. Tut | Lab | Stud | Off. Tut | Off. Hrs | Total |
| EPEG101 | Electrical Circuits (1)                           | 3            | 2             | 0       | 1.5      | 1.5 |      |          |          | 5     |
| EPEG102 | Principles of Energy Conversion                   | 2            | 2             | 0       | 0        | 0   |      |          | 1        | 3     |
| EPEG103 | Computer Applications in Electrical Power Systems | 3            | 2             | 0       | 1.5      | 1.5 |      |          |          | 5     |
| EECG118 | Electronics                                       | 3            | 2             | 2       | 0        | 0   |      |          |          | 4     |
| MTHG104 | Differential Equations                            | 3            | 2             | 2       | 0        | 0   |      |          |          | 4     |
| EMCG101 | Dynamics of Rigid bodies                          | 3            | 2             | 0       | 3        | 0   |      |          |          | 5     |
| INTG118 | Electric Materials                                | 2            | 2             | 0       | 0        | 0   |      |          |          | 2     |
| EPEG105 | Electrical Circuits (2)                           | 3            | 2             | 0       | 3        | 0   |      |          |          | 5     |
| EPEG106 | Logic Circuits and Microprocessors                | 3            | 2             | 0       | 3        | 0   |      |          |          | 5     |
| MTHG102 | Linear Algebra and Multivariable Integrals        | 3            | 2             | 2       | 0        | 0   |      |          |          | 4     |
| EPEG104 | Electromagnetic Fields                            | 4            | 3             | 2       | 0        | 0   |      |          |          | 5     |
| GENGxxx | Selected Topics 2 ***                             | 2            | 2             | 0       | 0        | 0   |      |          |          | 2     |

#### Level 200 Courses

| Code    | Name                              | Credit Hours | Contact Hours |         |          |     |      |          |          |       |
|---------|-----------------------------------|--------------|---------------|---------|----------|-----|------|----------|----------|-------|
|         |                                   |              | Lec           | Tut (2) | App. Tut | Lab | Stud | Off. Tut | Off. Hrs | Total |
| EPEG204 | Signals and Systems               | 3            | 2             | 0       | 1.5      | 1.5 |      |          |          | 5     |
| MEPG173 | Engineering Thermodynamics        | 3            | 2             |         | 3        |     |      |          |          | 5     |
| MDPG102 | Fundamentals of Mechanical Design | 2            | 2             | 0       | 0        | 0   |      |          |          | 2     |
| MTHG110 | Introduction to Numerical         | 2            | 1             | 0       | 3        | 0   |      |          |          | 4     |



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| Code    | Name                                 | Credit Hours | Contact Hours |         |          |     |      |          |          |       |
|---------|--------------------------------------|--------------|---------------|---------|----------|-----|------|----------|----------|-------|
|         |                                      |              | Lec           | Tut (2) | Ap p Tut | Lab | Stud | Off. Tut | Off. Hrs | Total |
|         | Analysis                             |              |               |         |          |     |      |          |          |       |
| CVEG118 | Civil Engineering                    | 2            | 2             | 0       | 0        | 0   |      |          | 1        | 2     |
| EPEG202 | Electrical Machines (1)              | 3            | 2             | 0       | 1.5      | 1.5 |      |          |          | 5     |
| CMPG218 | Data Structures                      | 2            | 1             | 0       | 3        | 0   |      |          |          | 4     |
| EPEG203 | Electric and Electronic Measurements | 3            | 2             | 0       | 1.5      | 1.5 |      |          |          | 5     |
| EPEG205 | Electrical Machines (2)              | 3            | 2             | 0       | 1.5      | 1.5 |      |          |          | 5     |
| MTHG113 | Probability and Statistics           | 2            | 1             | 0       | 3        | 0   |      |          |          | 4     |
| EPEG207 | Automatic Control Systems            | 3            | 2             | 0       | 1.5      | 1.5 |      |          |          | 5     |
| EPEG201 | Power Systems (1)                    | 3            | 2             | 0       | 1.5      | 1.5 |      |          |          | 5     |

### Level 300 Courses

| Code            | Name                              | Credit Hours | Contact Hours |         |          |     |       |          |          |       |
|-----------------|-----------------------------------|--------------|---------------|---------|----------|-----|-------|----------|----------|-------|
|                 |                                   |              | Lec           | Tut (2) | Ap p Tut | Lab | Stu d | Off. Tut | Off. Hrs | Total |
| EPEG301         | Power Systems (2)                 | 3            | 2             | 0       | 1.5      | 1.5 |       |          |          | 5     |
| EPEG303         | Electrical Machines (3)           | 3            | 2             | 0       | 1.5      | 1.5 |       |          |          | 5     |
| EPEG302         | Power Electronics (1)             | 3            | 2             | 0       | 1.5      | 1.5 |       |          |          | 5     |
| EPEG304         | Switchgear and Protection Systems | 3            | 2             | 0       | 1.5      | 1.5 |       |          |          | 5     |
| EPEG206         | Electrical Power Generation       | 3            | 2             | 0       | 1.5      | 1.5 |       |          |          | 5     |
| GENG351         | Selected Current Local Issues     | 1            | 1             | 0       | 0        | 0   |       |          |          | 1     |
| EPEG306         | Power Electronics (2)             | 3            | 2             | 0       | 1.5      | 1.5 |       |          |          | 5     |
| EPEG305         | High Voltage Engineering          | 3            | 2             | 0       | 1.5      | 1.5 |       |          |          | 5     |
| GENG1XX         | (UR) Restricted Elective-1        | 2            | 2             | 0       | 0        | 0   |       |          |          | 2     |
| EPEG307         | Digital Control Systems           | 3            | 2             | 0       | 3        | 0   |       |          |          | 5     |
| EECG318         | Electrical Communications Systems | 2            | 2             | 0       | 0        | 0   |       |          |          | 2     |
| GENG22X/24X/26X | (UR) Restricted Elective-2        | 2            | 2             | 0       | 0        | 0   |       |          |          | 2     |



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### Level 400 Courses

| Code            | Name                                         | Credit Hours | Contact Hours |         |          |     |       |          |          |       |
|-----------------|----------------------------------------------|--------------|---------------|---------|----------|-----|-------|----------|----------|-------|
|                 |                                              |              | Lec           | Tut (2) | Ap p Tut | Lab | Stu d | Off. Tut | Off. Hrs | Total |
| EPEG401         | Electrical Energy Utilization and Management | 3            | 2             | 2       | 0        | 0   |       |          |          | 4     |
| EPEG4XX         | Elective Course (1)                          | 3            | 2             | 0       | 1.5      | 1.5 |       |          |          | 5     |
| EPEG4XX         | Elective Course (2)                          | 3            | 2             | 0       | 1.5      | 1.5 |       |          |          | 5     |
| EPEG4XX         | Elective Course (3)                          | 3            | 2             | 0       | 1.5      | 1.5 |       |          |          | 5     |
| GENG21X/<br>23X | (UR) FREE ELECTIVE                           | 2            | 2             | 0       | 0        | 0   |       |          |          | 2     |
| EPEG480         | Graduation Project (1)                       | 1            |               |         |          |     |       |          |          | 0     |
| EPEG402         | Electrical Installations                     | 2            | 1             | 2       | 0        | 0   |       |          |          | 3     |
| EPE4GXX         | Elective Course (4)                          | 3            | 2             | 0       | 1.5      | 1.5 |       |          |          | 5     |
| EPE4GXX         | Elective Course (5)                          | 3            | 2             | 0       | 1.5      | 1.5 |       |          |          | 5     |
| EPE4GXX         | Elective Course (6)                          | 3            | 2             | 0       | 1.5      | 1.5 |       |          |          | 5     |
| EPEG481         | Graduation Project (2)                       | 3            |               |         |          |     |       |          |          | 0     |



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### COURSES CONTENTS توصيف المقررات

#### PROGRAM/DISCIPLINE REQUIREMENT

| Code     | Name/Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Credit Hours        | Contact Hours |         |                |     |      |              |          |       |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------|---------|----------------|-----|------|--------------|----------|-------|
|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                     | Lec           | Tut (2) | App Tut        | Lab | Stud | Off. Tut     | Off. Hrs | Total |
| EMCG101  | Dynamics of Rigid bodies                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 3                   | 2             |         | 3              |     |      |              |          | 5     |
|          | Pre-requisites: EMCG001                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Semester's Work: 60 |               |         | Final Exam: 40 |     |      | Exam Hrs.: 2 |          |       |
|          | Polar Moment of Inertia; Rigid Body Kinematics, Plane Motion, Types of Constrains and Applications; The Velocity Relation; The Acceleration Relation; The Instantaneous Center and The Wheel Motion; The Equations of Motion - Rectilinear Translation Motion - Rotational Motion; Curvilinear Translation Motion - General Plane Motion; Principle of Work and Kinetic Energy; Principle of Work and Mechanical Energy; Work done by Non-Conservative Forces (Constant Force, Constant Couple and Rolling Motion); Mathematical Modelling for Vibrations - Free Undamped Vibrations - Free Damped Vibrations; Forced Undamped Vibrations - Forced damped Vibrations |                     |               |         |                |     |      |              |          |       |
|          | Textbook <b>Mechanics for Engineers; Authors: R. C. Hibbeler and Kai Beng Yap; 14<sup>th</sup> edition; Pearson.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                     |               |         |                |     |      |              |          |       |
| CVEG118  | Civil Engineering                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2                   | 2             | 0       | 0              | 0   |      | 1            |          | 2     |
|          | Pre-requisites: EMCG001                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Semester's Work: 50 |               |         | Final Exam: 50 |     |      | Exam Hrs.: 2 |          |       |
|          | Fundamentals of Building Construction – Types of Buildings – Specifications of Materials Used in Buildings and Internal Partitions – Insulation against Heat, Moisture, Noise and Pollution – Architectural and Structural Drawings – Structure Analysis and Calculations.                                                                                                                                                                                                                                                                                                                                                                                           |                     |               |         |                |     |      |              |          |       |
| Textbook | <b>Sheng-Taur Mau, Sami Maalouf , Introduction to Civil Engineering, Revised 1st ed. Edition, 2014</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                     |               |         |                |     |      |              |          |       |
| CMPG218  | Data Structure                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2                   | 1             | 0       | 3              | 0   |      |              |          | 4     |
|          | Pre-requisites: AMTG005                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Semester's Work: 60 |               |         | Final Exam: 40 |     |      | Exam Hrs.: 2 |          |       |
|          | Data types and representation – file structures- data structures representation in storage media and memory allocation- linear lists - stacks - queues - memory allocation - trees - graphs - Hashing - searching, sorting algorithms and their analysis                                                                                                                                                                                                                                                                                                                                                                                                             |                     |               |         |                |     |      |              |          |       |
| Textbook | <b>Michael T. Goodrich, Roberto Tamassia, David M. Mount, Data Structures and</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                     |               |         |                |     |      |              |          |       |



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| Code           | Name/Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Credit Hours               | Contact Hours |          |                       |            |                     |          |          |          |  |
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|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                            | Lec           | Tut (2)  | App Tut               | Lab        | Stud                | Off. Tut | Off. Hrs | Total    |  |
|                | <b>Algorithms in C++, 2nd Edition, 2011</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                            |               |          |                       |            |                     |          |          |          |  |
| <b>EECG118</b> | <b>Electronics</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>3</b>                   | <b>2</b>      | <b>2</b> | <b>0</b>              | <b>0</b>   |                     |          |          | <b>4</b> |  |
|                | <b>Pre-requisites: PHYG001</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Semester's Work: 50</b> |               |          | <b>Final Exam: 50</b> |            | <b>Exam Hrs.: 2</b> |          |          |          |  |
|                | Basic Semiconductor Physics – PN Junction – Bipolar Junction Transistor -Field effect transistors– Mosfet - Basic Transistor Circuits – Operational Amplifiers (Op-Amp)– Basic Op-Amp Circuits – Active Filters. Basic Semiconductor Physics – PN Junction – Bipolar Junction Transistor -Field effect transistors– Mosfet - Basic Transistor Circuits – Operational Amplifiers (Op-Amp)– Basic Op-Amp Circuits – Active Filters. Use of filters in industrial power quality applications.                                                                                                                                                                                                                      |                            |               |          |                       |            |                     |          |          |          |  |
| Textbook       | <b>Paul Horowitz , Winfield Hill , The Art of Electronics, 3rd Edition, 2015</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                            |               |          |                       |            |                     |          |          |          |  |
| <b>EECG318</b> | <b>Electrical Communications Systems</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>2</b>                   | <b>2</b>      | <b>0</b> | <b>0</b>              | <b>0</b>   |                     |          |          | <b>2</b> |  |
|                | <b>Pre-requisites: MTHG110 + EPEG204</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Semester's Work: 50</b> |               |          | <b>Final Exam: 50</b> |            | <b>Exam Hrs.: 2</b> |          |          |          |  |
|                | Principles of Fiber Optics – SDH/SONET Fiber Optic Communication Systems – Fiber Optic Cables –Principles of Power Line Carrier (PLC) Communication Systems – PLC Equipment – Wireless Communications Overview – Basics of Wireless Networks – Wireless Systems. Applications to electric power industry.                                                                                                                                                                                                                                                                                                                                                                                                       |                            |               |          |                       |            |                     |          |          |          |  |
| Textbook       | <b>Mohammad A. Matin, Communication Systems for Electrical Engineers, Springer, 2018</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                            |               |          |                       |            |                     |          |          |          |  |
| <b>EPEG101</b> | <b>Electrical Circuits (1)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>3</b>                   | <b>2</b>      | <b>0</b> | <b>1.5</b>            | <b>1.5</b> |                     |          |          | <b>5</b> |  |
|                | <b>Pre-requisites: PHYG002+MTHG003</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Semester's Work: 60</b> |               |          | <b>Final Exam: 40</b> |            | <b>Exam Hrs.: 2</b> |          |          |          |  |
|                | Quantities of Electric Circuits: Potential Difference – Current – Energy – Power. Basic Laws of Electric Circuits: Ohm's Law - Kirchhoff' Laws. Elements and Components of Electric Circuits: Resistors – Capacitors – Inductors – Voltage and Current Sources. Methods and Theories of DC Circuits Analysis: Nodal Method – Mesh Method - Superposition Theorem – Thevenin's Theorem – Norton's Theorem. Dependent Sources: Dependent Voltage and Current Sources. Steady–State Analysis of AC Circuits: Sinusoidal Voltage and Current – Phasor Representation – AC Power. Switching Transients in DC and AC Circuits: First Order Circuits – Second Order Circuits. Applications to electric power industry. |                            |               |          |                       |            |                     |          |          |          |  |
| Textbook       | <b>Nilsson and Riedel, Electric Circuits, 2019, 11th edition</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                            |               |          |                       |            |                     |          |          |          |  |



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|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                     | Lec           | Tut (2) | App Tut        | Lab | Stud | Off. Tut     | Off. Hrs | Total |
| EPEG102  | Principles of Energy Conversion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2                   | 2             | 0       | 0              | 0   |      | 1            | 0        | 2     |
|          | Pre-requisites: PHYG002+MTHG003                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Semester's Work: 50 |               |         | Final Exam: 50 |     |      | Exam Hrs.: 2 |          |       |
|          | Introduction to the Electrical Grid – Different Types of Energy – Energy Measuring Units (Electrical –Mechanical – Chemical – Nuclear) – Energy Utilization and Load Growth Rates – Energy Reserve – Main Resources of Energy (Fossil – Nuclear Energy – Solar Energy – Wind Energy – Hydro Energy...etc.) – Production of Various Types of Energy (Thermal – Mechanical – Electrical ... etc.) – Basic Principles of Energy Storage – Batteries (Technologies – Types – Characteristics – Uses) – Energy Economics and Introduction to Energy Conservation. Applications to electrical power industry.                                                                                                                                                                                                                                                                                                                                                                                 |                     |               |         |                |     |      |              |          |       |
| Textbook | George G. Karady, Keith E. Holbert, Electrical Energy Conversion and Transport: An Interactive Computer-Based Approach, 2013, 2nd Edition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                     |               |         |                |     |      |              |          |       |
| EPEG104  | Electromagnetic Fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 4                   | 3             | 2       | 0              | 0   |      |              |          | 5     |
|          | Pre-requisites: PHYG002+MTHG110                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Semester's Work: 50 |               |         | Final Exam: 50 |     |      | Exam Hrs.: 3 |          |       |
|          | Electrostatic Fields: Fundamental postulates - Electric field intensity – Gauss's Law and its applications – Electric Potential – Dielectric Materials – Electric flux density – Boundary conditions – Capacitance and capacitors – Electric Energy and Forces<br>Magnetostatic Fields: Fundamental postulates – Magnetic flux density - Ampere's Law & Vector Magnetic Potential - Biot-Savart Law - Magnetic Dipoles & Scalar Magnetic Potential - Magnetization & Magnetic Materials - Boundary Conditions -Inductances and Inductors - Magnetic Energy - Magnetic Forces & Torque<br>Time varying Fields: Faraday's Law and its applications – Magnetic Circuits – Maxwell's Equations – Power and Poynting theorem – Electromagnetic Waves in different media – Quasi static magnetic fields applications: Induction Heating – Eddy power loss calculations in transformers – Skin effect and proximity effect – magnetic insulation. Applications to the electric power industry. |                     |               |         |                |     |      |              |          |       |
| Textbook | Andrew Zangwill, Modern Electrodynamics, 2013, 1st Edition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                     |               |         |                |     |      |              |          |       |
| EPEG103  | Computer Applications in Electrical Power Systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3                   | 2             | 0       | 1.5            | 1.5 |      |              |          | 5     |
|          | Pre-requisites: AMTG005                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Semester's Work: 60 |               |         | Final Exam: 40 |     |      | Exam Hrs.: 2 |          |       |
|          | Numbering Systems - Matrix operation and basic data manipulation using Matlab - Simulink representation of Electrical Circuits - Pointers, structure and Classes in high-                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                     |               |         |                |     |      |              |          |       |





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|          |                                                                                                                                                                                                                                                                                                                                                |                     | Lec           | Tut (2) | App Tut        | Lab | Stud | Off. Tut     | Off. Hrs | Total |  |
|          | level languages. MATLAB debugger. Line plots, surface plots, and animations..Matlab scripts. Different basic flow-control.                                                                                                                                                                                                                     |                     |               |         |                |     |      |              |          |       |  |
| Textbook | 1- Digital Design, 5th Edition, by MORRIS MANO, 2012<br>2-A. Gilat , MATLAB: An Introduction with Applications, 6th Ed., John Wiley & Sons, Inc., 2016.<br>3- C-How to program 7th edition, by : Paul Deitel and Harvey Deitel ,Pearson 2013                                                                                                   |                     |               |         |                |     |      |              |          |       |  |
| EPEG105  | Electrical Circuits (2)                                                                                                                                                                                                                                                                                                                        | 3                   | 2             | 0       | 3              | 0   |      |              |          | 5     |  |
|          | Pre-requisites: MTHG104+EPEG101                                                                                                                                                                                                                                                                                                                | Semester's Work: 60 |               |         | Final Exam: 40 |     |      | Exam Hrs.: 2 |          |       |  |
|          | Series and Parallel Resonance – Balanced and Unbalanced Three-Phase Circuits – Mutually Coupled Circuits – Application of Laplace - Transform in Circuit Analysis – Application of Fourier Transform in Circuit Analysis – Two-Port Circuits. Applications to the electric power industry.                                                     |                     |               |         |                |     |      |              |          |       |  |
| Textbook | Nilsson and Riedel, Electric Circuits, 2019, 11 <sup>th</sup> edition                                                                                                                                                                                                                                                                          |                     |               |         |                |     |      |              |          |       |  |
| EPEG106  | Logic Circuits and Microprocessors                                                                                                                                                                                                                                                                                                             | 3                   | 2             | 0       | 3              | 0   |      |              |          | 5     |  |
|          | Pre-requisites: EPEG103+EECG118                                                                                                                                                                                                                                                                                                                | Semester's Work: 60 |               |         | Final Exam: 40 |     |      | Exam Hrs.: 2 |          |       |  |
|          | Logic and Sequential Circuits – Counters and Timers – Memory – Microprocessor Structure –Microprocessor Programming – Microcontroller structure- Microcontroller programming- Interfacing Circuits – Microprocessor Based Instruments – Basics of Programmable Logic Controller. Applications to electric power industry, application project. |                     |               |         |                |     |      |              |          |       |  |
| Textbook | 1. Sepehr Naimi, Sarmad Naimi, and Muhammad Ali Mazidi. 2017. The AVR Microcontroller and Embedded Systems Using Assembly and C: Using Arduino Uno and Atmel Studio (2nd. ed.)<br>2. Digital Design, 5th Edition, by MORRIS MANO, 2012                                                                                                         |                     |               |         |                |     |      |              |          |       |  |
| EPEG201  | Power Systems (1)                                                                                                                                                                                                                                                                                                                              | 3                   | 2             | 0       | 1.5            | 1.5 |      |              |          | 5     |  |
|          | Pre-requisites: EPEG105+EPEG202                                                                                                                                                                                                                                                                                                                | Semester's Work: 60 |               |         | Final Exam: 40 |     |      | Exam Hrs.: 2 |          |       |  |
|          | Digital Design, 5th Edition, by MORRIS MANO, 2012<br>Transmission line Analysis: Parameters calculation – Performance analysis – A, B, C and D General Constants of Transmission Lines – Reactive Power and Voltage Control –                                                                                                                  |                     |               |         |                |     |      |              |          |       |  |





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|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |              | Lec                        | Tut (2)  | App Tut  | Lab                   | Stud       | Off. Tut            | Off. Hrs | Total |          |
|          | Compensated Transmission Lines – Transmission Line Design. Power Transformer Modeling: Three-Winding Ideal Transformer – Practical Three-Winding Transformer – Regulating Transformers for Voltage Magnitude and Angle – Per Unit Representation of Transformers with off nominal Turns Ratios. Symmetrical Components: Deriving Symmetrical Components for Voltages – Currents – Impedances – Power. Short Circuit Studies: Fault Analysis Model Simplification – Fault Calculation Assumptions – Fault Calculations – DC offset Calculations during Short Circuit. Applications to the electric power industry. Introduction to software packages used for short circuit calculations and transmission line modeling.                                                                                                                                                                                                                                       |              |                            |          |          |                       |            |                     |          |       |          |
| Textbook | <b>J. Duncan Glover, Mulukutla S. Sarma, Thomas Overbye,Power System Analysis and Design ,2017, 6th Edition</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |              |                            |          |          |                       |            |                     |          |       |          |
| EPEG202  | <b>Electrical Machines (1)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |              | <b>3</b>                   | <b>2</b> | <b>0</b> | <b>1.5</b>            | <b>1.5</b> |                     |          |       | <b>5</b> |
|          | <b>Pre-requisites: EPEG105+EPEG104</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |              | <b>Semester’s Work: 60</b> |          |          | <b>Final Exam: 40</b> |            | <b>Exam Hrs.: 2</b> |          |       |          |
|          | Transformers: Construction – theory of operation – equivalent circuit - Per-Unit System- Tests - Efficiency & Regulation - Special Transformers: auto transformers, three winding transformers, instrumentation transformers and pulse transformers – No-load current and inrush current - Three-phase transformers – Three phase transformer connection groups - Parallel operation – Transformer cooling – Transformer sizing<br>Principles of Electromechanical energy conversion: Energy conversion process – Mechanical Force and torque in electromagnetic systems – rotating machines<br>DC Machines: Construction – theory of operation of generators and motors – Magnetic characteristics – Classification of DC motors – Performance characteristics for DC Motors – Stability – Starting of DC motors – Speed Control of DC Motors – Braking of DC Motors –Applications: DC Motor Traction Systems - Troubleshooting and maintenance of DC motors |              |                            |          |          |                       |            |                     |          |       |          |
| Textbook | <b>Ion Boldea, Lucian N. Tutelea, Electric Machines, 2022, 2<sup>nd</sup> Edition</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |              |                            |          |          |                       |            |                     |          |       |          |
| EPEG203  | <b>Electric and Electronic Measurements</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |              | <b>3</b>                   | <b>2</b> | <b>0</b> | <b>1.5</b>            | <b>1.5</b> |                     |          |       | <b>5</b> |
|          | <b>Pre-requisites: EPEG101+EECG118</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |              | <b>Semester’s Work: 60</b> |          |          | <b>Final Exam: 40</b> |            | <b>Exam Hrs.: 2</b> |          |       |          |
|          | Conventional Measurement Instruments: DC Measuring Instruments – AC Measuring Instruments. Electronic Measuring Instruments: Oscilloscopes – Signal Generators – Signal Analysers. Thermal Sensors – Mechanical Sensors – Optical Sensors – Ultrasonic                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |              |                            |          |          |                       |            |                     |          |       |          |



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|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |              | Lec                        | Tut (2)  | App Tut  | Lab                   | Stud       | Off. Tut            | Off. Hrs | Total |          |
|          | Sensors – Analog Signal Conditioning – Digital Signal Conditioning – Principles of Digital Measurements – Microprocessor Based Measurement Instrumentation. Application project.                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |              |                            |          |          |                       |            |                     |          |       |          |
| Textbook | <b>Prithwiraj Purkait, Electrical and Electronics Measurements and Instrumentation, McGraw Hill Education (India), 2013</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |              |                            |          |          |                       |            |                     |          |       |          |
| EPEG204  | <b>Signals and Systems</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |              | <b>3</b>                   | <b>2</b> | <b>0</b> | <b>1.5</b>            | <b>1.5</b> |                     |          |       | <b>5</b> |
|          | <b>Pre-requisites: MTHG104+EPEG101</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |              | <b>Semester's Work: 60</b> |          |          | <b>Final Exam: 40</b> |            | <b>Exam Hrs.: 2</b> |          |       |          |
|          | Signal Manipulation – System Properties – Modeling of Dynamic Systems – Convolution – Fourier -Transform – Transfer Functions – Block Diagram Simplification – Time Response – Stability –Z-Transform – Filters. Applications to electric power industry.                                                                                                                                                                                                                                                                                                                                                                                                 |              |                            |          |          |                       |            |                     |          |       |          |
| Textbook | <b>Alan Oppenheim, Alan Willsky, Signals and Systems, 2013</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |              |                            |          |          |                       |            |                     |          |       |          |
| EPEG205  | <b>Electrical Machines (2)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |              | <b>3</b>                   | <b>2</b> | <b>0</b> | <b>1.5</b>            | <b>1.5</b> |                     |          |       | <b>5</b> |
|          | <b>Pre-requisites: EPEG202</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |              | <b>Semester's Work: 60</b> |          |          | <b>Final Exam: 40</b> |            | <b>Exam Hrs.: 2</b> |          |       |          |
|          | AC Machines Fundamentals: Types of fields: Pulsating fields and Rotating fields – MMF distribution in AC Machines - EMF induced in AC machines – Torque production in AC Machines                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |              |                            |          |          |                       |            |                     |          |       |          |
|          | Three-phase induction motors: Construction - Principle of Operation - Equivalent Circuit – Power flow - Torque-speed characteristic – Testing –Performance, Loading & Stability - Induction Machine Modes of Operation - Starting of Induction motors: DOL Starting, star-delta starting, auto-transformer starting, soft starting, deep bar motors and double cage motors - Speed Control of Induction Motors - Braking of Induction Motors: Plugging, Dynamic braking and Regenerative Braking –Motor Selection for different applications – Induction motors in electrical traction applications - Troubleshooting and maintenance of induction motors |              |                            |          |          |                       |            |                     |          |       |          |
| Textbook | Two-phase induction motor: Balanced operation – unbalanced operation – equivalent circuit – power flow – application: AC servo motor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |              |                            |          |          |                       |            |                     |          |       |          |
|          | Single-phase induction motor: Principle of operation – Double revolving field theory – Equivalent circuit – Performance & Power flow – Types: Split-phase motor, capacitor-start motor, capacitor-run motor, capacitor start-capacitor-run motor and shaded pole motor – Tests – Selection of the starting resistor and capacitor – Reversing – Speed Control                                                                                                                                                                                                                                                                                             |              |                            |          |          |                       |            |                     |          |       |          |
| Textbook | <b>Ion Boldea, Lucian N. Tutelea, Electric Machines, 2022, 2<sup>nd</sup> Edition</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |              |                            |          |          |                       |            |                     |          |       |          |



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|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                     | Lec           | Tut (2) | App Tut        | Lab | Stud         | Off. Tut | Off. Hrs | Total |
| EPEG206  | Electrical Power Generation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 3                   | 2             | 0       | 1.5            | 1.5 |              |          |          | 5     |
|          | Pre-requisites: EPEG101+MEPG173                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Semester's Work: 60 |               |         | Final Exam: 40 |     | Exam Hrs.: 2 |          |          |       |
|          | Load Curves – Introduction to Different Types of Power Plants (Thermal, Hydro, ....etc.) – Economics of Power Plants and Tariffs –Thermal Power Plants – Different Electrical and mechanical Components in Power Plants – Economic Operation of Power Plants and Connection between Plants (Loads and Frequency Relations – Governor System – Synchronization with Grid – Unit Commitment – Distributed Generation – Safety Measures in Electrical Power Plants – Protection of Power Plants – Electrical Energy Storage Systems Design – Introduction to Smart Grids.- Required Pre studies before building power plant<br>Practical Applications: Role of Egyptian NCC- Site visit (term project) |                     |               |         |                |     |              |          |          |       |
|          | Textbook: S. Sivanagaraju, D. Srilatha, Generation and Utilization of Electrical Energy, 2010                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                     |               |         |                |     |              |          |          |       |
| EPEG207  | Automatic Control Systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 3                   | 2             | 0       | 1.5            | 1.5 |              |          |          | 5     |
|          | Pre-requisites: EPEG204                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Semester's Work: 60 |               |         | Final Exam: 40 |     | Exam Hrs.: 2 |          |          |       |
|          | Linear Continuous Systems: Transient and Steady State Response Analysis – Root Locus Analysis – Control Systems Design using the Root Locus Methods – Control Systems Design by the Frequency Response using Bode Diagrams – PID Controllers Design and Tuning - Analysis of Control Systems in State-Space – Pole Placement – Design of Servo Systems – State Observers – Design of Control Systems with Observers. Applications to electric power industry.                                                                                                                                                                                                                                       |                     |               |         |                |     |              |          |          |       |
| Textbook | Norman S. Nise, Control Systems Engineering, 8th Edition, 2019                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                     |               |         |                |     |              |          |          |       |
| EPEG301  | Power Systems (2)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 3                   | 2             | 0       | 1.5            | 1.5 |              |          |          | 5     |
|          | Pre-requisites: EPEG201+MTHG110                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Semester's Work: 60 |               |         | Final Exam: 40 |     | Exam Hrs.: 2 |          |          |       |
|          | Network Modeling: Formation of Bus Admittance Matrix by Inspection and by Singular Transformation – Power System Graph – Incidence Matrices. Load Flow Study: Importance and Problem Formulation – Bus Loading Equations –Network Performance Equations – Line Flows and Power Loss Calculations – Different Solution Methods – Applications using Software Packages. Short Circuit Studies using Matrices: Representation of Power Systems under Faults – Fault Matrices for Different Fault Types –Calculations of Fault Variables – Applications. Economic Operation of Power Systems: Input-output Characteristic – Distribution of Loads among Generators –Transmission                        |                     |               |         |                |     |              |          |          |       |



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## BYLAWS 2023

### Bachelor of Science Degree Credit Hours System



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| Code     | Name/Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Credit Hours               | Contact Hours |          |                       |            |      |                     |          |          |  |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|---------------|----------|-----------------------|------------|------|---------------------|----------|----------|--|
|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                            | Lec           | Tut (2)  | App Tut               | Lab        | Stud | Off. Tut            | Off. Hrs | Total    |  |
|          | System Active Power Loss Matrix – Distribution of Loads among Generators with Transmission Losses – Iterative Solution of Economic Dispatch Problem. Power System Stability: Swing equation – single and multi-machine stability problem– Equal area criteria, Power System Reliability. Applications and a project.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                            |               |          |                       |            |      |                     |          |          |  |
| Textbook | <b>J. Duncan Glover, Mulukutla S. Sarma, Thomas Overbye, Power System Analysis and Design ,2017, 6th Edition</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                            |               |          |                       |            |      |                     |          |          |  |
| EPEG302  | <b>Power Electronics (1)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>3</b>                   | <b>2</b>      | <b>0</b> | <b>1.5</b>            | <b>1.5</b> |      |                     |          | <b>5</b> |  |
|          | <b>Pre-requisites: EECG118+EPE105</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Semester's Work: 60</b> |               |          | <b>Final Exam: 40</b> |            |      | <b>Exam Hrs.: 2</b> |          |          |  |
|          | Power Semiconductor Devices: (Power Diodes – Power Bipolar Junction Transistor – Thyristors) Basic Structure – Output Characteristics. Rectifier Circuits: Analysis and Design of Single-Phase Rectifiers (Uncontrolled –Fully Controlled – Semi-Controlled Rectifiers) – Analysis and Design of Three-Phase Rectifiers (Uncontrolled – Fully Controlled –Semi-Controlled Rectifiers) Thyristor Firing Circuits: Analog Firing Circuits – Digital Firing Circuits DC/DC Converters: Basic Switching Converters (The Buck Regulator – The Boost Regulator – The Buck/Boost Regulator). Different configuration of DC Drive applications - Starting techniques of DC machines and basics of speed control - sizing and design of dc drives - DC choppers for speed regulation of dc machines - Different braking and reversing techniques of dc machines - different speed control algorithms. Applications to electric power industry. |                            |               |          |                       |            |      |                     |          |          |  |
| Textbook | <b>Andrzej M. Trzynadlowski, Introduction to Modern Power Electronics, 3rd Edition, 2015.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                            |               |          |                       |            |      |                     |          |          |  |
| EPEG303  | <b>Electrical Machines (3)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>3</b>                   | <b>2</b>      | <b>0</b> | <b>1.5</b>            | <b>1.5</b> |      |                     |          | <b>5</b> |  |
|          | <b>Pre-requisites: EPEG205</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Semester's Work: 60</b> |               |          | <b>Final Exam: 40</b> |            |      | <b>Exam Hrs.: 2</b> |          |          |  |
|          | Synchronous Generators: Construction – MMF distribution in turbo and salient machines – EMF induced - Principle of Operation – Space and Time Phasor Diagram - Characteristics & Tests: Open circuit, short circuit and zero power factor – Synchronous Reactance – Voltage Regulation – Speed Regulation - Generator Modes of Operation: Stand-alone, parallel operation and infinite bus-bar operation – Synchronization – Transient Electrodynamics performance and Stability – Synchronizing Power and Torque – Generator Rating and Power Capability Curve – Two Reaction Theory - Generator Selection and Specifications - Troubleshooting and maintenance of Synchronous generators.<br>Synchronous Motors: Modes of operation, Starting and Synchronous condensers                                                                                                                                                            |                            |               |          |                       |            |      |                     |          |          |  |



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| Code     | Name/Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Credit Hours        | Contact Hours |         |                |     |              |          |          |       |  |
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|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                     | Lec           | Tut (2) | App Tut        | Lab | Stud         | Off. Tut | Off. Hrs | Total |  |
|          | Special Synchronous Motors: Three-phase Synchronous Reluctance Motors – Brushless DC Motors - Variable Reluctance Motors.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                     |               |         |                |     |              |          |          |       |  |
| Textbook | Ion Boldea, Lucian N. Tutelea, Electric Machines, 2022, 2 <sup>nd</sup> Edition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                     |               |         |                |     |              |          |          |       |  |
| EPEG304  | Switchgear and Protection Systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 3                   | 2             | 0       | 1.5            | 1.5 |              |          |          | 5     |  |
|          | Pre-requisites: EPEG201+EPEG203                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Semester's Work: 60 |               |         | Final Exam: 40 |     | Exam Hrs.: 2 |          |          |       |  |
|          | Abnormalities in Electric Power Systems (Over-Load – Short-Circuit – Over-Voltage) – Short circuit calculations - Relays principles (Electromagnetic – Thermal – Static) – Protection systems: (Over-Load Protection – Distance Protection) – Protection against earth faults - Introduction to Digital Protection – Protection System Coordination – Electric Power Apparatus Protection: Generators – Motors – Transformers – Transmission and Distribution Lines – Types of Circuit Breakers - Specifications of High Voltage circuit breaker - Overvoltage protection. Current and voltage transformers. Applications to electric power industry. |                     |               |         |                |     |              |          |          |       |  |
| Textbook | Power System Protection and Switchgear, McGraw-Hill Education, 2010. Switchgear & Protection Paperback English by J.B Gupta - 2019                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                     |               |         |                |     |              |          |          |       |  |
| EPEG305  | High Voltage Engineering                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 3                   | 2             | 0       | 1.5            | 1.5 |              |          |          | 5     |  |
|          | Pre-requisites: EPEG203                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Semester's Work: 60 |               |         | Final Exam: 40 |     | Exam Hrs.: 2 |          |          |       |  |
|          | High Voltage generations Alternative current, direct current and impulse current – High Voltage measurements direct and indirect methods – Uses of high voltage for testing and power transfer – Comparison between high voltage waveforms – Earthing system – Overvoltage and the traveling waves along high voltage transmission system and underground cables – Classification and construction of high voltage underground cables. Applications to electric power industry.                                                                                                                                                                       |                     |               |         |                |     |              |          |          |       |  |
| Textbook | 1. “High Voltage Engineering” M S Naidu and V Kamaraju. 5th edition 2013<br>2. High voltage engineering fundamentals, technology and applications, Andreas Kuchler 5th edition 2016                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                     |               |         |                |     |              |          |          |       |  |
| EPEG306  | Power Electronics (2)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3                   | 2             | 0       | 1.5            | 1.5 |              |          |          | 5     |  |
|          | Pre-requisites: EPEG302                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Semester's Work: 60 |               |         | Final Exam: 40 |     | Exam Hrs.: 2 |          |          |       |  |
|          | Power Semiconductor Devices (Power MOSFET): Basic Structure, Output Characteristics, Switching Characteristics and Losses - Inverters: Full Bridge Inverter Structure, Square Wave Output, Fourier Series Analysis and THD, Amplitude and Harmonics Control, Half Bridge Inverter Structure – Pulse Width Modulated Output                                                                                                                                                                                                                                                                                                                            |                     |               |         |                |     |              |          |          |       |  |





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|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|---------------|----------|-----------------------|----------|------|---------------------|----------|----------|--|
|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                            | Lec           | Tut (2)  | App Tut               | Lab      | Stud | Off. Tut            | Off. Hrs | Total    |  |
|          | Techniques – PWM Harmonics Analysis – Multi-level Inverter Topologies and Operation – Three-Phase Inverter Structure – Switching Techniques in Three-Phase Inverter: 180 Conduction and PWM - AC Machines Drives: Structure and Modelling of AC Machines – Power Electronics Circuits for AC Machine Drives – AC Drives Modelling and Control - Machine Drives Practical Considerations. Applications to electric power industry. Application project.                                                                                                                                                                                                                                                                     |                            |               |          |                       |          |      |                     |          |          |  |
| Textbook | <b>Andrzej M. Trzynadlowski, Introduction to Modern Power Electronics, 3rd Edition, 2015.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                            |               |          |                       |          |      |                     |          |          |  |
| EPEG307  | <b>Digital Control Systems</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>3</b>                   | <b>2</b>      | <b>0</b> | <b>3</b>              | <b>0</b> |      |                     |          | <b>5</b> |  |
|          | <b>Pre-requisites: EPEG207</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Semester's Work: 60</b> |               |          | <b>Final Exam: 40</b> |          |      | <b>Exam Hrs.: 2</b> |          |          |  |
|          | Sampling Theorem – Transfer Functions in Z Domain – Block Diagram Manipulation – Root Locus in Z - Domain – Direct Digital Control Design – PID Controllers – Transformation Techniques – Design by Emulation – Introduction to Least-Squares Identification – Discrete-Time State-Space Models – Controllability – Observability – Discrete-Time State Feedback – Discrete-Time Output Feedback. Applications to electric power industry.                                                                                                                                                                                                                                                                                 |                            |               |          |                       |          |      |                     |          |          |  |
| Textbook | <b>Phillips, Nagle, and Chakraborty, Digital Control System Analysis &amp; Design, 4th Edition, 2015</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                            |               |          |                       |          |      |                     |          |          |  |
| EPEG401  | <b>Electrical Energy Utilization and Management</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>3</b>                   | <b>2</b>      | <b>2</b> | <b>0</b>              | <b>0</b> |      |                     |          | <b>4</b> |  |
|          | <b>Pre-requisites: EPEG205+EPEG306</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Semester's Work: 50</b> |               |          | <b>Final Exam: 50</b> |          |      | <b>Exam Hrs.: 2</b> |          |          |  |
|          | Electric Lamps Technology and Harmonics Suppression – Electric Heating, Welding and Electric Furnaces – Electrochemical Processes (Electroplating – Materials Extraction...etc.) – Electrical Transportation Systems (Lifts and Escalators – Electrical Vehicles – Electric Traction...etc.) – Power Factor Correction – Energy Efficient Technologies in Electrical Systems – Buildings Energy Efficient Management – Energy Conservation Opportunities and Auditing for Electrical Industrial Systems (On-site Generators – Lighting –Compressed air – Fans and Pumped Water – HVAC – Chilled Water) – Introduction to Different Industrial Thermal System Utility Types, Its Energy Management and Efficient Operation. |                            |               |          |                       |          |      |                     |          |          |  |
| Textbook | <b>S. Sivanagaraju, D. Srilatha, Generation and Utilization of Electrical Energy, 2010</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                            |               |          |                       |          |      |                     |          |          |  |



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|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------|---------|----------------|-----|------|--------------|----------|-------|
|         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                     | Lec           | Tut (2) | App Tut        | Lab | Stud | Off. Tut     | Off. Hrs | Total |
| EPEG402 | Electrical Installations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2                   | 1             | 2       | 0              | 0   |      |              |          | 5     |
|         | Pre-requisites: EPEG304+EPEG205                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Semester's Work: 50 |               |         | Final Exam: 50 |     |      | Exam Hrs.: 2 |          |       |
|         | Residential and Industrial Load Estimation (Lifts– Air Conditioning– Pumps, etc ) – Design of Different Lighting Systems (Internal – External –Street lighting, ....etc.) – Residential and Industrial Distribution Systems – Design of Distribution Boards and Feeding Systems – Design of Branch Circuits (Cable Sizing – Circuit Breaker Selection – Internal Wiring) – Estimation – Emergency Systems (ATS, UPS Specification and Sizing) –Earthing Systems in residential buildings– Light Current Systems (Fire Alarm – CCTV –..... etc.) – Standard Specifications and Tests of Electric Equipment. Use of ETAP and ecodial in a complete design (term project). |                     |               |         |                |     |      |              |          |       |
|         | Textbook                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                     |               |         |                |     |      |              |          |       |
| EPEG280 | Summer Internship                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1                   | 0             | 0       | 0              | 0   |      |              |          | 0     |
|         | Pre-requisites: 55 Credit hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Semester's Work: -  |               |         | Final Exam: -  |     |      | Exam Hrs.: - |          |       |
|         | It is a credit-based module oriented towards training on industrial establishments relevant to the program and specialization. Training lasts for total of 6-8 weeks. Adding to the mentor/trainer's in the industrial establishment formal report on the student's performance during training, the student submits a training certificate, a formal report and presentation to be evaluated by a panel of members appointed from the program. The course is graded as Pass/Fail grade-system.                                                                                                                                                                         |                     |               |         |                |     |      |              |          |       |
|         | Textbook                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                     |               |         |                |     |      |              |          |       |
| EPEG480 | Graduation Project (1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1                   |               |         |                |     |      |              |          | 0     |
|         | Pre-requisites: 120 Credit hours + all sophomore courses                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Semester's Work: -  |               |         | Final Exam: -  |     |      | Exam Hrs.: - |          |       |
|         | Students undertake a major project as part of the program. The aim of the project is to provide the students, who work in groups, with an opportunity to implement appropriate concepts and techniques to a particular design. Students are required to select and research the expected project to be designed and implemented in the following course Graduation Project-2. The student should give an oral presentation to be approved. The course is graded as Pass/Fail grade-system.                                                                                                                                                                              |                     |               |         |                |     |      |              |          |       |
|         | Textbook                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                     |               |         |                |     |      |              |          |       |



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|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|---------------|----------|-----------------------|----------|------|---------------------|----------|----------|
|                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                           |                            | Lec           | Tut (2)  | App Tut               | Lab      | Stud | Off. Tut            | Off. Hrs | Total    |
| <b>EPEG481</b><br> | <b>Graduation Project (2)</b>                                                                                                                                                                                                                                                                                                                                                                                                             | <b>3</b>                   |               |          |                       |          |      |                     |          | <b>0</b> |
|                                                                                                    | <b>Pre-requisites: EPEG480</b>                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Semester's Work: -</b>  |               |          | <b>Final Exam: -</b>  |          |      | <b>Exam Hrs.: -</b> |          |          |
|                                                                                                    | All students undertake a major project as part of the program. The aim of the project is to provide the students, who work in groups, with an opportunity to implement the appropriate concepts and techniques to a particular design. A dissertation on the project is submitted on which the student is examined orally.                                                                                                                |                            |               |          |                       |          |      |                     |          |          |
| <b>INTG118</b>                                                                                     | <b>Electrical Materials</b>                                                                                                                                                                                                                                                                                                                                                                                                               | <b>2</b>                   | <b>2</b>      | <b>0</b> | <b>0</b>              | <b>0</b> |      |                     |          | <b>2</b> |
|                                                                                                    | <b>Pre-requisites: PHYG001</b>                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Semester's Work: 50</b> |               |          | <b>Final Exam: 50</b> |          |      | <b>Exam Hrs.: 2</b> |          |          |
|                                                                                                    | Atomic Structure and Interatomic Bonding – Solid materials crystal structure – X-ray diffraction and their applications - Electrical Properties (Electronic and Ionic Conduction, Energy Band Structures in Solids, Bonding Models, Electron Mobility, Resistivity) – Electrical material classifications from electrical properties (Alloys, Semiconductor, Dielectric, conductor) Magnetic Properties – Magnetic materials applications |                            |               |          |                       |          |      |                     |          |          |
| Textbook                                                                                           | <b>Advanced Electrical and Electronics Materials: Processes and Applications, 2015, K.M. Gupta, Nishu Gupta.</b>                                                                                                                                                                                                                                                                                                                          |                            |               |          |                       |          |      |                     |          |          |
| <b>MDPG102</b>                                                                                     | <b>Fundamentals of Mechanical Design</b>                                                                                                                                                                                                                                                                                                                                                                                                  | <b>2</b>                   | <b>1</b>      | <b>0</b> | <b>3</b>              | <b>0</b> |      |                     |          | <b>4</b> |
|                                                                                                    | <b>Pre-requisites: INTG001</b>                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Semester work: 60</b>   |               |          | <b>Final Exam: 40</b> |          |      | <b>Exam Hrs.: 2</b> |          |          |
|                                                                                                    | Design methodology – Factors of safety – Machine parts – Fasteners – Welding – Shafts and axles – Wedges, keys and cotters – Bearings – Gears – Belts – Seals. Applications to the electric power industry.                                                                                                                                                                                                                               |                            |               |          |                       |          |      |                     |          |          |
| Textbook                                                                                           | <b>Shigley's Mechanical Engineering Design [10 ed.] McGraw-Hill Series in Mechanical Engineering, 2014</b>                                                                                                                                                                                                                                                                                                                                |                            |               |          |                       |          |      |                     |          |          |
| <b>MEPG173</b>                                                                                     | <b>Engineering Thermodynamics</b>                                                                                                                                                                                                                                                                                                                                                                                                         | <b>3</b>                   | <b>2</b>      |          | <b>3</b>              |          |      |                     |          | <b>5</b> |
|                                                                                                    | <b>Pre-requisites: INTG001</b>                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Semester work: 60</b>   |               |          | <b>Final Exam: 40</b> |          |      | <b>Exam Hrs.: 2</b> |          |          |
|                                                                                                    | Introduction – Definitions - Pure Substances - Heat and Work for a closed and open system - The first law of Thermodynamics - Second law of Thermodynamics, Carnot principle - Rankine cycle – Air standard Cycles – Brayton Cycle – Introduction to heat transfer ( conduction – convection – radiation). Applications to electric power industry.                                                                                       |                            |               |          |                       |          |      |                     |          |          |





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|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---------------|---------|---------|-----|------|----------|----------|-------|
|          |                                                                                                                                                                                                                                                 |              | Lec           | Tut (2) | App Tut | Lab | Stud | Off. Tut | Off. Hrs | Total |
| Textbook | “Thermodynamics: An Engineering Approach”, ninth Edition, 2019, By: Y. Cengel and M. Boles.<br>“Fundamentals of Engineering Thermodynamics”, 9th Edition, 2018, By: Michael J. Moran, Howard N. Shapiro, Daisie D. Boettner, Margaret B. Bailey |              |               |         |         |     |      |          |          |       |

## ELECTIVE COURSES

| ELECTIVE COURSES |                                                                                                                                                                                                                                                                              |                     |               |         |                |     |              |         |           |       |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------|---------|----------------|-----|--------------|---------|-----------|-------|
| Code             | Name/ Content                                                                                                                                                                                                                                                                | Credit              | Contact Hours |         |                |     |              |         |           |       |
|                  |                                                                                                                                                                                                                                                                              |                     | Lec           | Tut (2) | App Tut        | Lab | Stud         | Off. Hr | Off. Tut. | Total |
| EPEG411          | Electrical Power System Planning                                                                                                                                                                                                                                             | 3                   | 2             | 0       | 1.5            | 1.5 |              |         |           | 5     |
|                  | Pre-requisites: EPEG301 + AA Approval                                                                                                                                                                                                                                        | Semester's Work: 60 |               |         | Final Exam: 40 |     | Exam Hrs.: 2 |         |           |       |
|                  | Load Estimation – Load Forecasting – Optimal Mathematical Techniques for Planning – Reliability. Practical Applications: estimation of reliability index in distribution networks-reliability studies required before building a generation power plant                      |                     |               |         |                |     |              |         |           |       |
| Textbook         | Hossein Seifi, Mohammad Sadegh Sepasian, Electric Power System Planning: Issues, Algorithms and Solutions, Springer, 2011                                                                                                                                                    |                     |               |         |                |     |              |         |           |       |
| EPEG412          | Advanced Studies in Electrical Power Systems                                                                                                                                                                                                                                 | 3                   | 2             | 0       | 1.5            | 1.5 |              |         |           | 5     |
|                  | Pre-requisites: EPEG301+ AA Approval                                                                                                                                                                                                                                         | Semester's Work: 60 |               |         | Final Exam: 40 |     | Exam Hrs.: 2 |         |           |       |
|                  | Power Factor Improvement – Flexible AC Transmission Systems (FACTS). Shunt FACTS devices (STATCOM, SVC) – Series FACTS devices (TCSC, SSSC) and Applications – HVDC Representation in Power System Studies – Loss Reduction and Voltage Control. Optimal power flow problem. |                     |               |         |                |     |              |         |           |       |
| Textbook         | José M. Maza-Ortega and Antonio Gómez-Expósito, HVDC/FACTS for Grid Services in Electric Power Systems, 2020                                                                                                                                                                 |                     |               |         |                |     |              |         |           |       |
| EPEG413          | Electrical Power Quality                                                                                                                                                                                                                                                     | 3                   | 2             | 0       | 1.5            | 1.5 |              |         |           | 5     |
|                  | Pre-requisites: EPEG301 + AA Approval                                                                                                                                                                                                                                        | Semester's Work: 60 |               |         | Final Exam: 40 |     | Exam Hrs.: 2 |         |           |       |



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|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------|---------|----------------|-----|------|--------------|-----------|-------|
| Code             | Name/ Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Credit              | Contact Hours |         |                |     |      |              |           |       |
|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                     | Lec           | Tut (2) | App Tut        | Lab | Stud | Off. Hr      | Off. Tut. | Total |
|                  | Introduction to Power Quality – Classification of Power Quality Variations (Transients – Short Duration Voltage Variations – Long Duration Voltage Variations – Voltage Imbalance – Voltage Fluctuations (Flicker) – Waveform Distortion) – Overview on Different Power Quality Issues – Effect of Power Quality on Different Electrical Equipment (Lamps – Cables – Transformers – Motor Drives...etc.) – Harmonic Sources – Harmonic Indices (Harmonic Factor – Total Harmonic Distortion –...etc.) – Harmonic Mitigation and Basics of Passive, Active and Hybrid Filters – Voltage Sag and Swells (Causes, Indices and Mitigation Methods).<br>Practical Applications: Deign of filters used in PQ improvements using software packages |                     |               |         |                |     |      |              |           |       |
| Textbook         | Surya Santoso , Fundamentals Of Electric Power Quality, 2010                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                     |               |         |                |     |      |              |           |       |
| EPEG414          | Electrical Substations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 3                   | 2             | 0       | 1.5            | 1.5 |      |              |           | 5     |
|                  | Pre-requisites: EPEG301 + AA Approval                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Semester's Work: 60 |               |         | Final Exam: 40 |     |      | Exam Hrs.: 2 |           |       |
|                  | Introduction to Substations and their Main Functions and Types – Different Substation Components and Equipment – Transformers Substation and Bus-Bar Distribution System in Power Plants – Interlocking Systems for Substation Equipment – Substations Protection Schemes (Feeders – Transformers – Bus-bars – Capacitor Banks... etc.) – Monitoring and Operation Schemes in the substations – Substations Auxiliary Schemes (DC Systems, AC Systems, Pumps, etc.) – Substations Components Design and Sizing – Feeding Systems – Maintenance, Testing and Commissioning (Transformers – Protection Relays – DC Batteries – Cables – Bus-bars – Switchgear...etc.).                                                                        |                     |               |         |                |     |      |              |           |       |
| Textbook         | John D. McDonald, Electric Power Substations Engineering (The Electrical Engineering Handbook) 3rd Edition, 2012                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                     |               |         |                |     |      |              |           |       |
| EPEG415          | Renewable Energy Systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 3                   | 2             | 0       | 1.5            | 1.5 |      |              |           | 5     |
|                  | Pre-requisites: EPEG301 + AA Approval                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Semester's Work: 60 |               |         | Final Exam: 40 |     |      | Exam Hrs.: 2 |           |       |



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### ELECTIVE COURSES

| ELECTIVE COURSES |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                            |               |          |                       |            |      |                     |           |          |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|---------------|----------|-----------------------|------------|------|---------------------|-----------|----------|
| Code             | Name/ Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Credit                     | Contact Hours |          |                       |            |      |                     |           |          |
|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                            | Lec           | Tut (2)  | App Tut               | Lab        | Stud | Off. Hr             | Off. Tut. | Total    |
|                  | Introduction about different types of renewable energy resources- Off Grid Solar (PV) standalone systems (design and selection of different system components)- Off Grid Solar (PV) water pumping system for irrigation (design and selection of different system components)- Introduction to Renewable Energy economics and feasibility studies- Introduction to design software of Solar (PV) systems.<br>Practical application: sizing of PV system (PV cells, battery, inverter, etc)                                                                                                                                                                                                                                  |                            |               |          |                       |            |      |                     |           |          |
| Textbook         | <b>Radian Belu, Renewable Energy Systems: Fundamentals and Source Characteristics, 2022</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                            |               |          |                       |            |      |                     |           |          |
| EPEG416          | <b>Renewable Energy Sources Grid Connected Systems</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>3</b>                   | <b>2</b>      | <b>0</b> | <b>1.5</b>            | <b>1.5</b> |      |                     |           | <b>5</b> |
|                  | <b>Pre-requisites: EPEG301 + AA Approval</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Semester's Work: 60</b> |               |          | <b>Final Exam: 40</b> |            |      | <b>Exam Hrs.: 2</b> |           |          |
|                  | Wind Energy systems (selection and design of different system components)- Grid Connected Systems Requirements, Components and Regulations for PV and Wind Systems – Grid Synchronization Techniques for Power Conditioning Units (PCUs) – Grid Current Regulation, Grid Interconnection Filter Design – Types and Components of Grid Connected Power Conditioning Units – PCU Specifications, Reliability and Design Aspects – Operation of PCU during Abnormal Conditions and Grid Faults – Islanding Detection – Anti-Islanding Methods – Relaying Systems for Renewable Energy Grid Integration – Testing and Commissioning of Grid Connected Systems – Economic Evaluation for Grid Connected Renewable Energy Sources |                            |               |          |                       |            |      |                     |           |          |
| Textbook         | <b>Ahmad Azar, Nashwa Kamal, Renewable Energy Systems: Modelling, Optimization and Control, 1st Edition, 2021</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                            |               |          |                       |            |      |                     |           |          |
| EPEG417          | <b>Demand Side Management</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>3</b>                   | <b>2</b>      | <b>0</b> | <b>1.5</b>            | <b>1.5</b> |      |                     |           | <b>5</b> |
|                  | <b>Pre-requisites: EPEG301 + AA Approval</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Semester's Work: 60</b> |               |          | <b>Final Exam: 40</b> |            |      | <b>Exam Hrs.: 2</b> |           |          |
|                  | Load Characteristics – Consumer Factors – DSM Objectives and Techniques – Tariffs – EMS Functions and Configurations – Cogeneration applications – Distributed Generation – Highly Efficient Lighting – Energy Efficient Motors. Demand side management alternatives. Power tools and applications.                                                                                                                                                                                                                                                                                                                                                                                                                         |                            |               |          |                       |            |      |                     |           |          |



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| ELECTIVE COURSES |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                     |               |         |                |     |      |              |           |       |
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| Code             | Name/ Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Credit              | Contact Hours |         |                |     |      |              |           |       |
|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                     | Lec           | Tut (2) | App Tut        | Lab | Stud | Off. Hr      | Off. Tut. | Total |
| Textbook         | Prachi Gupta, Demand Side Management: An Approach To Peak Load Smoothing, 2013                                                                                                                                                                                                                                                                                                                                                                                                                                          |                     |               |         |                |     |      |              |           |       |
|                  | Cybersecurity Applied to the National Grid                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 3                   | 2             | 0       | 1.5            | 1.5 |      |              |           | 5     |
| EPEG418          | Pre-requisites: EPEG301 + AA Approval                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Semester's Work: 60 |               |         | Final Exam: 40 |     |      | Exam Hrs.: 2 |           |       |
|                  | Definitions: Cyberspace – Cybersecurity – External Attacks – Internal Incidences. ICS: PLC – DCS - SCADA. Malware: Types – Protection – Treatment. Attack: Surface – Vector – Type. Grounding and Cybersecurity. International and Local Accidents: Malware – Type – Damage – Cost – ...etc. Vulnerabilities: Generation – Transmission – Distribution – Customer Premise. Supply Chain: Manufacturers – Vendors – Updates – Upgrades – ...etc. Standards: CIP, IEEE, IEC, Local – ...etc. Mini Projects and Lab Demos. |                     |               |         |                |     |      |              |           |       |
| Textbook         | Charles J. Brooks, Christopher Grow, and Donald Short, Cybersecurity Essentials, 2018                                                                                                                                                                                                                                                                                                                                                                                                                                   |                     |               |         |                |     |      |              |           |       |
|                  | Data Communication for Instrumentation and Control                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 3                   | 2             | 0       | 1.5            | 1.5 |      |              |           | 5     |
| EPEG419          | Pre-requisites EPEG307 + AA Approval                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Semester's Work: 60 |               |         | Final Exam: 40 |     |      | Exam Hrs.: 2 |           |       |
|                  | Basic Principles – Different Media for Data Communication (Ethernet, Fibre Optic and Wireless) – Physical Layer Standards – Open Systems Interconnection Model – Industrial Protocols (TCP/IP, Modbus, DNP3, and IEC 60870) – Industrial Networks (Industrial Ethernet – DeviceNet – Profibus...etc.)                                                                                                                                                                                                                   |                     |               |         |                |     |      |              |           |       |
| Textbook         | Larry Lawrence Larry M. Thompson , Tim Shaw, Industrial Data Communications: Fifth Edition (2016)                                                                                                                                                                                                                                                                                                                                                                                                                       |                     |               |         |                |     |      |              |           |       |
|                  | Special Topics in Automatic Control                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 3                   | 2             | 0       | 1.5            | 1.5 |      |              |           | 5     |
| EPEG420          | Pre-requisites: EPEG307 + AA Approval                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Semester's Work: 60 |               |         | Final Exam: 40 |     |      | Exam Hrs.: 2 |           |       |
|                  | System Identification - time-invariant system identification - Time varying system identification - Model validation - Model validation techniques. Invariance principle - Sliding mode control - Variable structure control - Chattering Analysis - Sliding mode                                                                                                                                                                                                                                                       |                     |               |         |                |     |      |              |           |       |



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| ELECTIVE COURSES |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                     |               |         |                |     |              |         |           |       |
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| Code             | Name/ Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Credit              | Contact Hours |         |                |     |              |         |           |       |
|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                     | Lec           | Tut (2) | App Tut        | Lab | Stud         | Off. Hr | Off. Tut. | Total |
|                  | observers. Output feedback sliding node control.                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                     |               |         |                |     |              |         |           |       |
| Textbook         | <b>Computer Controlled Systems: Theory and Design, 3rd Edition, 2011, by K. Astrom and B. Wittenmark (Dover Books on Electrical engineering)</b><br><b>Variable-Structure Systems and Sliding-Mode Control: From Theory to Practice (Studies in Systems, Decision and Control, 271) 1st ed. 2020 Edition</b>                                                                                                                                                                                              |                     |               |         |                |     |              |         |           |       |
| EPEG421          | Intelligent Control                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3                   | 2             | 0       | 1.5            | 1.5 |              |         |           | 5     |
|                  | Pre-requisites: EPEG307 + AA Approval                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Semester's Work: 60 |               |         | Final Exam: 40 |     | Exam Hrs.: 2 |         |           |       |
|                  | Expert and Fuzzy Control – Mathematics of Fuzzy control - Stability analysis of Fuzzy system - Fuzzy rule base - Evolutionary Fuzzy control. Logical computations with neurons - Neural networks layers - Neural Network Control. Applications to the electric power industry. Application project.                                                                                                                                                                                                       |                     |               |         |                |     |              |         |           |       |
| Textbook         | <b>Anastasios Dounis, Intelligent Control in Energy Systems, August 2019</b>                                                                                                                                                                                                                                                                                                                                                                                                                              |                     |               |         |                |     |              |         |           |       |
| EPEG422          | Industrial Control System Components                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 3                   | 2             | 0       | 1.5            | 1.5 |              |         |           | 5     |
|                  | Pre-requisites: EPEG307 + AA Approval                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Semester's Work: 60 |               |         | Final Exam: 40 |     | Exam Hrs.: 2 |         |           |       |
|                  | Sensors and transducers, Position sensors, Effort and Force sensors, Level and flow sensors, Pressure and temperature sensors, Digital transducers, Actuators, System identification, Introduction to Robotics, Robot kinematics & inverse kinematics. Piping & Instrumentation Diagrams (P&ID), Tuning feedback loops, Cascade Control, Override Control, Feedforward Control, Smith predictor, Dahlin & IMC, Implementation of Industrial Controllers, Multivariable control, Model Predictive Control. |                     |               |         |                |     |              |         |           |       |
| Textbook         | <b>Tuning of industrial control systems, third edition, publisher: ISA. 2015.</b><br><b>Principles and practice of automatic process control, 3rd edition 2006</b>                                                                                                                                                                                                                                                                                                                                        |                     |               |         |                |     |              |         |           |       |
| EPEG423          | Computer Controlled Systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 3                   | 2             | 0       | 1.5            | 1.5 |              |         |           | 5     |
|                  | Pre-requisites: EPEG307 + AA Approval                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Semester's Work: 60 |               |         | Final Exam: 40 |     | Exam Hrs.: 2 |         |           |       |
|                  | Components of Computer Controlled Systems – Distributed Control Systems (DCS): Hierarchy – Process Units – Local Control Units – Data Communication Networks – Monitoring and Supervisory Control– SCADA Software and Hardware Features – Basic                                                                                                                                                                                                                                                           |                     |               |         |                |     |              |         |           |       |



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| ELECTIVE COURSES |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                            |               |         |                       |     |      |                     |           |       |
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| Code             | Name/ Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Credit                     | Contact Hours |         |                       |     |      |                     |           |       |
|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                            | Lec           | Tut (2) | App Tut               | Lab | Stud | Off. Hr             | Off. Tut. | Total |
|                  | Design and Engineering Requirements for Computer Control Systems – Applications.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                            |               |         |                       |     |      |                     |           |       |
| Textbook         | <b>Industrial Automation Technologies by Chanchal Dey</b> , Sunit Kumar Sen · CRC press, 2020                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                            |               |         |                       |     |      |                     |           |       |
| EPEG424          | <b>Selected Topics in Electrical Machines and Transformers</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 3                          | 2             | 0       | 1.5                   | 1.5 |      |                     |           | 5     |
|                  | <b>Pre-requisites: EPEG303 + AA Approval</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Semester’s Work: 60</b> |               |         | <b>Final Exam: 40</b> |     |      | <b>Exam Hrs.: 2</b> |           |       |
|                  | Transformers: Inrush Currents; Causes and Reduction Methods – K-Factor Transformers for Nonlinear Loads – Effects of Unbalanced Operation under Normal and Short Circuit Conditions – Special Designs to Eliminate Zero Sequence Components.<br>Electrical Motors: Electrical Motors Fed from Unbalanced and/or Non-Sinusoidal Supply: Sources of Unbalance and Harmonics – Motor Equivalent Circuits – Temperature Rise Calculations – Derating Rules – Single Phasing Phenomenon – Operation of Three-Phase Motors from Single-Phase Supply – Design of Phase Balancers.<br>Analysis of Start-up and Re-switching Transients – Effect of the Different Types of Voltage Sag – Methods of Reducing Inrush. |                            |               |         |                       |     |      |                     |           |       |
| Textbook         | <b>Ion Boldea, Lucian N. Tutelea, Electric Machines, 2022, 2nd Edition</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                            |               |         |                       |     |      |                     |           |       |
| EPEG425          | <b>Applications of Computers in Electrical Machine Design</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 3                          | 2             | 0       | 1.5                   | 1.5 |      |                     |           | 5     |
|                  | <b>Pre-requisites: EPEG303 + AA Approval</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Semester’s Work: 60</b> |               |         | <b>Final Exam: 40</b> |     |      | <b>Exam Hrs.: 2</b> |           |       |
|                  | Basic Principles of Electrical Machines and Transformers Design<br>Induction Motors: Output equation of Induction motor – Main dimensions – Length of air gap- Rules for selecting rotor slots of squirrel cage machines – Design of rotor bars & end rings – Design of wound rotor – Methods of cooling – Magnetic circuit-Magnetizing current.<br>Transformers: Output Equations – Main Dimensions - KVA output for single and three phase transformers – Window space factor – Overall dimensions – Operating characteristics –No load current – Temperature rise in Transformers – Design of Tank - Methods of cooling of Transformers - Computer-aided electrical machines design                      |                            |               |         |                       |     |      |                     |           |       |





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| ELECTIVE COURSES |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                |               |         |                |     |              |         |           |       |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------|---------|----------------|-----|--------------|---------|-----------|-------|
| Code             | Name/ Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Credit                                                                                                         | Contact Hours |         |                |     |              |         |           |       |
|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                | Lec           | Tut (2) | App Tut        | Lab | Stud         | Off. Hr | Off. Tut. | Total |
| Textbook         | Alexander Gray, Electrical Machine Design: The Design and Specification of Direct and Alternating Current Machinery–2018                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                |               |         |                |     |              |         |           |       |
| EPEG426          | Electric Drive Systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3                                                                                                              | 2             | 0       | 1.5            | 1.5 |              |         |           | 5     |
|                  | Pre-requisites: EPEG303 + AA Approval                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Semester's Work: 60                                                                                            |               |         | Final Exam: 40 |     | Exam Hrs.: 2 |         |           |       |
|                  | DC Drives: DC Motors Speed Control (Field Control – Armature Control) – Types of Rectifiers used in DC Drives (Single-Phase and Three-Phase Rectifiers) – Types of DC Choppers used in DC Drives – Applications of DC Drives (Traction, Rolling Mills).<br>AC Drives: Adjustable Speed Drive (ASD) for AC Motors – AC Drives Topologies – AC Drives Modelling and Control – Types of AC Drives Control (Open-loop, Scalar Control, Vector Control, Sensor-less Control, Direct Torque Control) – Some Applications of ASD (Pumps – Hoists – Centrifuges) - Energy Saving Concept in Fans and Pump Drive Systems<br>PM Synchronous Motor and Brushless DC Motor Drives. |                                                                                                                |               |         |                |     |              |         |           |       |
|                  | Textbook                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Austin Hughes, Bill Drury, Electric Motors and Drives: Fundamentals, Types and Applications, 2019, 5th Edition |               |         |                |     |              |         |           |       |
| EPEG427          | Advanced Drive Systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3                                                                                                              | 2             | 0       | 1.5            | 1.5 |              |         |           | 5     |
|                  | Pre-requisites: EPEG426 + AA Approval                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Semester's Work: 60                                                                                            |               |         | Final Exam: 40 |     | Exam Hrs.: 2 |         |           |       |
|                  | Introduction to Reference Frame Theory of Electrical Machines: Voltage Differential Equations and Mechanical Equations of Electrical Machines in the ABC Frame – Two-phase Transformation of Electrical Machines – Commutator Transformation of Electrical Machines – Park's Transformation – Voltage Differential Equations and Mechanical Equations of Electrical Machines in the DQ Frame. Modern Trends in Adjustable AC Drive Systems: Scalar Control – Vector Control – Sensor and Sensor-less Controlled Drives                                                                                                                                                 |                                                                                                                |               |         |                |     |              |         |           |       |
|                  | Textbook                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Austin Hughes, Bill Drury, Electric Motors and Drives: Fundamentals, Types and Applications, 2019, 5th Edition |               |         |                |     |              |         |           |       |
| EPEG428          | Electric Traction and Mobility Systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3                                                                                                              | 2             | 0       | 1.5            | 1.5 |              |         |           | 5     |
|                  | Pre-requisites: EPE302+ EPEG303+ AA Approval                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Semester's Work: 60                                                                                            |               |         | Final Exam: 40 |     | Exam Hrs.: 2 |         |           |       |





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|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|---------------|---------|-----------------------|-----|------|---------------------|-----------|-------|
| Code             | Name/ Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Credit                     | Contact Hours |         |                       |     |      |                     |           |       |
|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                            | Lec           | Tut (2) | App Tut               | Lab | Stud | Off. Hr             | Off. Tut. | Total |
|                  | Evolution of electric traction systems and electric vehicles – DC Traction Systems – AC Traction Systems – Selection of Motors for Electric Traction and Mobility systems – DC Traction Motors - AC Traction motors- AC Commutator Machines (Principle of operation – equivalent circuit – performance) - linear Induction Motor - Special Motors for Traction and Mobility systems: construction, theory of operation, performance – Control systems of Traction and Mobility motors: speed control and braking                                                    |                            |               |         |                       |     |      |                     |           |       |
| Textbook         | <b>Andreas Steimel, Electric Traction: Motive Power and Energy Supply, 2014, 2nd Edition</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                            |               |         |                       |     |      |                     |           |       |
| EPEG429          | <b>Electrical Generators for Renewable Energy Applications</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 3                          | 2             | 0       | 1.5                   | 1.5 |      |                     |           | 5     |
|                  | <b>Pre-requisites: EPEG303 + AA Approval</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Semester’s Work: 60</b> |               |         | <b>Final Exam: 40</b> |     |      | <b>Exam Hrs.: 2</b> |           |       |
|                  | Wind energy conversion principles - General introduction - Types and classification of wind energy conversion systems - Power, torque and speed characteristics – Maximum power point tracking of wind turbines. Induction generators: electrical characteristics - slip - speed torque characteristics - Self excited induction generator - Constant speed Induction generators - Variable speed Induction generators - Doubly fed Induction generators. Permanent magnet synchronous generators. Advanced Modeling of electrical generators - Generators control. |                            |               |         |                       |     |      |                     |           |       |
| Textbook         | <b>Ion Boldea, Electric Generators Handbook - Two Volume Set, 2016, 2nd edition.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                            |               |         |                       |     |      |                     |           |       |
| EPEG430          | <b>Selected Topics in High Voltage Engineering</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 3                          | 2             | 0       | 1.5                   | 1.5 |      |                     |           | 5     |
|                  | <b>Pre-requisites: EPEG305 + AA Approval</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Semester’s Work: 60</b> |               |         | <b>Final Exam: 40</b> |     |      | <b>Exam Hrs.: 2</b> |           |       |
|                  | Gas, liquid and solid insulators - electrical breakdown in gasses - destructive and non-destructive tests - ionic corona - leakage currents in high voltage insulators - electrical breakdown in liquid insulators - high voltage cables - protection against transient voltages - electrical insulators for transmission lines                                                                                                                                                                                                                                     |                            |               |         |                       |     |      |                     |           |       |
| Textbook         | <b>“High Voltage Engineering” M S Naidu and V Kamaraju. 5th edition 2013.”</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                            |               |         |                       |     |      |                     |           |       |



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### ELECTIVE COURSES

| ELECTIVE COURSES |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                     |               |         |                |     |              |         |           |       |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------|---------|----------------|-----|--------------|---------|-----------|-------|
| Code             | Name/ Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Credit              | Contact Hours |         |                |     |              |         |           |       |
|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                     | Lec           | Tut (2) | App Tut        | Lab | Stud         | Off. Hr | Off. Tut. | Total |
| EPEG431          | High Voltage Engineering Applications                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 3                   | 2             | 0       | 1.5            | 1.5 |              |         |           | 5     |
|                  | Pre-requisites: EPEG305 + AA Approval                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Semester's Work: 60 |               |         | Final Exam: 40 |     | Exam Hrs.: 2 |         |           |       |
|                  | Corona interference with radio and television waves – Biological effect for the electromagnetic waves – electrostatic precipitators – electrostatic charges in industrial – electrostatic painting and x-ray - Electrostatic dry imaging.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                     |               |         |                |     |              |         |           |       |
| Textbook         | “High Voltage Engineering” M S Naidu and V Kamaraju. 5th edition 2013.”                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                     |               |         |                |     |              |         |           |       |
| EPEG432          | High Voltage Direct Current Transmission                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 3                   | 2             | 0       | 1.5            | 1.5 |              |         |           | 5     |
|                  | Pre-requisites: EPEG305 + AA Approval                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Semester's Work: 60 |               |         | Final Exam: 40 |     | Exam Hrs.: 2 |         |           |       |
|                  | Historical background on HVDC - Advantages and reasons for using HVDC - Power electronics components used in HVDC - Types of HVDC converter stations, components, and working principle - Types of HVDC Cables and electrostatic field calculations - HVDC circuit breaker types and working principles - HVDC Power flow control - Mitigation of harmonics arising from HVDC converter stations. Coordination of digital overcurrent protection - Modern digital protection for power transformers - Modern mechanical protection for power transformers - Modern digital protection for generators and busbars - Protection of multi-end transmission lines - Protection schemes for distance protection - Modern digital protection for motors, capacitors, and induction reactors - Automatic re-closing of lines Transport. |                     |               |         |                |     |              |         |           |       |
| Textbook         | High Voltage Direct Current Transmission: Converters, Systems and DC Grids, 2019, Second Edition, Dragan Jovcic.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                     |               |         |                |     |              |         |           |       |
| EPEG433          | Advanced Topics in protection Systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 3                   | 2             | 0       | 1.5            | 1.5 |              |         |           | 5     |
|                  | Pre-requisites: EPEG304 + AA Approval                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Semester's Work: 60 |               |         | Final Exam: 40 |     | Exam Hrs.: 2 |         |           |       |
|                  | Coordination of digital overcurrent protection - Modern digital protection for power transformers - Modern mechanical protection for power transformers - Modern digital protection for generators and busbars - Protection of multi-end transmission lines - Protection schemes for distance protection - Modern digital protection for motors, capacitors, and induction reactors - Automatic re-closing of lines Transport.                                                                                                                                                                                                                                                                                                                                                                                                   |                     |               |         |                |     |              |         |           |       |



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### ELECTIVE COURSES

| ELECTIVE COURSES |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                     |               |         |                |     |      |              |           |       |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------|---------|----------------|-----|------|--------------|-----------|-------|
| Code             | Name/ Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Credit              | Contact Hours |         |                |     |      |              |           |       |
|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                     | Lec           | Tut (2) | App Tut        | Lab | Stud | Off. Hr      | Off. Tut. | Total |
| Textbook         | Power System Protection and Switchgear, McGraw-Hill Education, 2010.<br>Switchgear & Protection Paperback English by J.B Gupta - 2019.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                     |               |         |                |     |      |              |           |       |
| EPEG434          | DC Circuit Breakers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 3                   | 2             | 0       | 1.5            | 1.5 |      |              |           | 5     |
|                  | Pre-requisites: AA Approval                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Semester's Work: 60 |               |         | Final Exam: 40 |     |      | Exam Hrs.: 2 |           |       |
|                  | Principles of DC circuit breakers - types of DC circuit breakers - DC circuit breaker specifications - current transformers – Optical current transformers- DC current transformers - voltage transformers - circuit breaker tests                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                     |               |         |                |     |      |              |           |       |
| Textbook         | J.B Gupta, Switchgear & Protection, 2019.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                     |               |         |                |     |      |              |           |       |
| EPEG435          | Power Electronics Applications in Renewable Energy Systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 3                   | 2             | 0       | 1.5            | 1.5 |      |              |           | 5     |
|                  | Pre-requisites: EPEG306 + AA Approval                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Semester's Work: 60 |               |         | Final Exam: 40 |     |      | Exam Hrs.: 2 |           |       |
|                  | Renewable Energy Systems: Wind Power Conversion – Fuel Cell Power Conversion – Photovoltaic Cell. Wind Power Conversion: Basic Control Variables for Wind Turbines – Maximum Power Point Tracking Techniques – Generators Technology for Wind Turbines– Power Converters for Wind Turbines – Controls and Grid integration Requirements for Wind Turbines.Photovoltaic Energy Conversion Systems: Power Curves and Maximum Power Point of PV Systems– Grid-Connected PV System Configurations – Control of Grid-Connected PV Systems – Recent Developments in Multilevel Inverter-Based PV Systems. Bulk Energy Storage: Pumped Storage in Hydro Power Plant – Battery Storage – Flywheel (FW) Storage – Superconducting Magnet Energy Storage. |                     |               |         |                |     |      |              |           |       |
| Textbook         | Gilsoo Jang, Power Electronics Applications in Renewable Energy Systems, 2021                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                     |               |         |                |     |      |              |           |       |
| EPEG436          | Power Electronics Applications in Power Systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3                   | 2             | 0       | 1.5            | 1.5 |      |              |           | 5     |
|                  | Pre-requisites: EPEG306 + AA Approval                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Semester's Work: 60 |               |         | Final Exam: 40 |     |      | Exam Hrs.: 2 |           |       |
|                  | Thyristor Based FACTS Devices: Conventional Reactive Power Compensators – Saturated Reactor (SR) – Thyristor-Controlled Reactor (TCR) – Thyristor-Controlled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                     |               |         |                |     |      |              |           |       |



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### ELECTIVE COURSES

| ELECTIVE COURSES |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                            |               |          |                       |            |      |                     |           |          |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|---------------|----------|-----------------------|------------|------|---------------------|-----------|----------|
| Code             | Name/ Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Credit                     | Contact Hours |          |                       |            |      |                     |           |          |
|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                            | Lec           | Tut (2)  | App Tut               | Lab        | Stud | Off. Hr             | Off. Tut. | Total    |
|                  | Transformer (TCT) – Fixed Capacitor/Thyristor-Controlled Reactor (FC–TCR) – Thyristor-Switched Capacitor (TSC) – Thyristor-Controlled Series Capacitor (TCSC) – Back-to-Back HVDC Conversion System.<br>Voltage Source Inverters: Static Synchronous Compensator (STATCOM) – Static Synchronous Series Compensator (SSSC) – Shunt Active Power Filters – Series Active Power Filter – Hybrid Active Power Filter – Back-to-Back HVDC VSC Conversion System.                                                                                                                                                                                                                                                                                                                                                                       |                            |               |          |                       |            |      |                     |           |          |
| Textbook         | <b>Neeraj Vyas Saifullah Khalid, Applications of Power Electronics in Power System, 2010</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                            |               |          |                       |            |      |                     |           |          |
| EPEG437          | <b>Embedded Systems Applications in Power Electronics</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>3</b>                   | <b>2</b>      | <b>0</b> | <b>1.5</b>            | <b>1.5</b> |      |                     |           | <b>5</b> |
|                  | <b>Pre-requisites: EPEG306 + AA Approval</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Semester's Work: 60</b> |               |          | <b>Final Exam: 40</b> |            |      | <b>Exam Hrs.: 2</b> |           |          |
|                  | Embedded Systems Design: Common Processor Architectures – Basic Microcontroller Hardware Systems – Interfacing Techniques – Microcontroller programming Using C-Language – Microcontroller programming Using MATLAB and Simulink – Software Development, Debugging and Testing.<br>Power Electronics Applications: Basic Analog Circuits for Digital Applications – Microcontroller-based Phase Control Triggering circuits (AC/DC Rectifiers and AC/AC Voltage Controllers) – Microcontroller-based PWM Control Drive Circuits (DC/DC Converters and DC/AC Converters) – Analog to Digital Conversion – Voltage Sensing and Measurement – Current Sensors and Current Measurement – Speed Sensors and Speed Measurement – Common Isolation Techniques – Implementation of Common Digital Controllers and Filters – Mini-project. |                            |               |          |                       |            |      |                     |           |          |
|                  | Textbook                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                            |               |          |                       |            |      |                     |           |          |
|                  | <b>Ahmet Bindal, Electronics for Embedded Systems, Springer, 2017</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                            |               |          |                       |            |      |                     |           |          |
| EPEG438          | <b>Power Electronics for electric vehicles</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>3</b>                   | <b>2</b>      | <b>0</b> | <b>1.5</b>            | <b>1.5</b> |      |                     |           | <b>5</b> |
|                  | <b>Pre-requisites: EPEG306 + AA Approval</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Semester's Work: 60</b> |               |          | <b>Final Exam: 40</b> |            |      | <b>Exam Hrs.: 2</b> |           |          |
|                  | Drivetrain Architecture and dynamics of hybrid and electric vehicles - rating and sizing of drivetrain components - Analysis, modeling, simulations and design considerations for: Battery systems, battery management electronics, Bidirectional dc-dc converters, Inverters and ac motor drives, Battery chargers - complete system modeling.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                            |               |          |                       |            |      |                     |           |          |



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### ELECTIVE COURSES

| ELECTIVE COURSES |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                        |               |         |                |     |              |         |           |       |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|---------------|---------|----------------|-----|--------------|---------|-----------|-------|
| Code             | Name/ Content                                                                                                                                                                                                                                                                                                                                                                                                                                           | Credit                 | Contact Hours |         |                |     |              |         |           |       |
|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                        | Lec           | Tut (2) | App Tut        | Lab | Stud         | Off. Hr | Off. Tut. | Total |
| Textbook         | L. Ashok Kumar, S. Albert Alexander, Power Converters for Electric Vehicles 1st Edition, 2020                                                                                                                                                                                                                                                                                                                                                           |                        |               |         |                |     |              |         |           |       |
| EPEG439          | Operations Research                                                                                                                                                                                                                                                                                                                                                                                                                                     | 3                      | 2             | 0       | 1.5            | 1.5 |              |         |           | 5     |
|                  | Pre-requisites:<br>MTHG110+ AA Approval                                                                                                                                                                                                                                                                                                                                                                                                                 | Semester's Work:<br>60 |               |         | Final Exam: 40 |     | Exam Hrs.: 2 |         |           |       |
|                  | Introduction – Linear Programming – Transport Model – Assignment Model – Non-Linear Programming – Methods of Decision Making – Applications.                                                                                                                                                                                                                                                                                                            |                        |               |         |                |     |              |         |           |       |
| Textbook         | Hamdy A. Taha, "Operations Research: An Introduction", 10th edition, Pearson Education Limited, England, 2017                                                                                                                                                                                                                                                                                                                                           |                        |               |         |                |     |              |         |           |       |
| EPEG440          | Electrical Project Management                                                                                                                                                                                                                                                                                                                                                                                                                           | 3                      | 2             | 0       | 1.5            | 1.5 |              |         |           | 5     |
|                  | Pre-requisites:<br>MTHG110+ AA Approval                                                                                                                                                                                                                                                                                                                                                                                                                 | Semester's Work:<br>60 |               |         | Final Exam: 40 |     | Exam Hrs.: 2 |         |           |       |
|                  | Bidding Analysis – pricing analysis - Bill of Quantities – direct and indirect expenses - Risk assessment – row material planning - cost estimation methods- Finance Cost – labor cost – equipment cost – sub-contractors cost - Site Overheads – general overheads - accounting records - pay back estimate– spare parts management – scheduling the BID – overhead check list – vendors – Conceptual Design – codes and specs – technical pre-studies |                        |               |         |                |     |              |         |           |       |